

Java NumberFormatException – Lab Report

1. What is NumberFormatException?

NumberFormatException is a runtime exception in Java that occurs when a program tries to convert a String into a numeric value, but the String does not contain a valid number. For example, **Integer.parseInt("ABC")** cannot convert "ABC" into an integer, so Java throws NumberFormatException.

2. Objective

To understand and handle NumberFormatException in Java by converting a String into an integer using Integer.parseInt() and handling the invalid input with try-catch.

3. Task Description

Write a Java program that attempts to convert the String value "ABC" into an integer. Since "ABC" is not a valid numeric string, NumberFormatException is generated. Handle the exception using a catch block and display a meaningful error message.

4. Step-wise Execution

1. Start the Java program.
2. Declare a String variable value and assign it the value "ABC".
3. Use Integer.parseInt(value) to convert the String into an integer.
4. Java checks whether the String represents a valid integer.
5. Because "ABC" is not a valid integer, NumberFormatException is thrown.
6. Program control moves from the try block to the catch block.
7. The catch block displays: "Exception: Invalid number format".
8. The program ends normally after handling the exception.

5. Algorithm

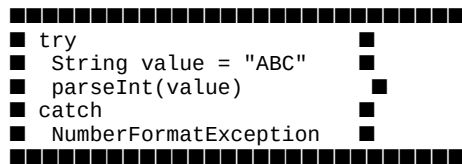
Step	Algorithm
1	Start
2	Set value = "ABC"
3	Try to convert value to integer using Integer.parseInt(value)
4	If conversion is successful, display the integer
5	If NumberFormatException occurs, catch it
6	Display "Exception: Invalid number format"
7	Stop

6. UML Diagram

```

classDiagram
    class NumberFormatDemo {
        + main(args: String[])
    }

```



Flow: Start → String "ABC" → Integer.parseInt() → NumberFormatException → catch block → display error message → End

7. Program Code

```
class NumberFormatDemo {
    public static void main(String[] args) {
        try {
            String value = "ABC";
            int number = Integer.parseInt(value);
            System.out.println(number);
        } catch (NumberFormatException e) {
            System.out.println("Exception: Invalid number format");
        }
    }
}
```

8. Code Explanation

- **try:** Contains the statements that may generate an exception.
- **String value = "ABC";** Stores a non-numeric string.
- **Integer.parseInt(value):** Attempts to convert the String into an int.
- **NumberFormatException:** Thrown because "ABC" cannot be converted into an integer.
- **catch (NumberFormatException e):** Catches and handles the exception.
- **System.out.println(...):** Displays a user-friendly error message.

9. Expected Output

Exception: Invalid number format

10. Output Screenshot

Java Program Output

Exception: Invalid number format

A screenshot of the Java program's output, showing the text "Exception: Invalid number format" in a monospaced font, enclosed in a rectangular box.

11. Result

The Java program successfully demonstrates NumberFormatException handling. The invalid string "ABC" causes the exception, and the catch block handles it without terminating the program abnormally.